The Climate Credit Channel: Physical Shocks and Monetary Policy Transmission in South Africa

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Roadmap

1. Credit channel
2. Climate channel
3. Monetary policy surprises: structural and high frequency
4. Climate shocks
5. Empirical specification
6. Results
7. Conclusion

Monetary policy transmission: beyond the interest rate channel

- Traditional interest rate mechanism: policy rate $\rightarrow$ real rate $\rightarrow$ cost of capital (Mishkin 1995)
- Empirical puzzles that this mechanism struggles with (Bernanke and Gertler 1995):
  - Spending responses are too large and too fast relative to estimated cost-of-capital elasticities
  - Heterogeneity: firms respond differently to the same policy shock, particularly depending on their financial position and access to credit.
- The credit channel is not a separate channel: it is an **amplification and propagation mechanism** for the conventional interest rate effect
- The object that does the amplifying is the **external finance premium**

The external finance premium (EFP)

Definition. The wedge between the cost of funds raised externally (debt or equity) and the opportunity cost of internal funds:

$$\text{EFP}_t \equiv r_t^{ext} - r_t^{int} > 0$$

- Why is it positive? Asymmetric information, agency costs, costly state verification and monitoring (Bernanke and Gertler 1989; Hubbard 1998)
- The external finance premium falls when borrowers have stronger balance sheets and more collateral relative to their assets, because lenders see them as less risky (Bernanke et al. 1996)

Lending rates: the price of credit

- Textbook pass-through from the policy rate r^p to bank lending rates r^L is complete and fast:

$$r^L_{b,t} \;=\; \alpha_b + \beta_b\, r^p_t + \varepsilon_{b,t}$$

- Full passthrough case, $\beta_b = 1$
- But in practice pass-through is incomplete, sluggish and heterogeneous across banks, and depends on bank balance sheets (Altavilla et al. 2020)
- Pass-through is faster when the policy move is anticipated (Kleimeier and Sander 2006)
- Lending rates embed the premium: $r^L = r^p + \text{EFP}$

The Credit Channel

- Financial frictions amplify monetary policy transmission through two channels:
- **Balance sheet channel:** policy affects borrowers' net worth, cash flow and collateral.
  - A policy tightening weakens borrower balance sheets through three routes (Bernanke and Gertler 1995; Kiyotaki and Moore 1997):
    - **Cash-flow effect:** higher debt service burden
    - **Collateral effect:** lower asset prices reduce the value of collateral
    - **Demand effect:** lower sales and revenues
- Weaker balance sheets $\rightarrow$ higher EFP $\rightarrow$ lower spending, with a "flight to quality" away from weak borrowers (Bernanke et al. 1996; Gertler and Gilchrist 1994)
- The same logic applies to households via wealth and debt service burdens (Slacalek et al. 2020)
- **Bank lending channel:** policy affects banks' capacity or willingness to supply loans.
- Monetary tightening can reduce bank funding and constrain loan supply.
- Strongest for small, illiquid banks (Kashyap and Stein 2000); composition of external finance shifts (Kashyap et al. 1993)
- In South Africa the bank lending channel is operative (Pirozhkova and Viegi 2025)

The credit channel is not fully understood

- The **magnitude** of the EFP response to policy varies across time, borrowers and products; even its originators flagged that the quantitative importance is unsettled (Bernanke and Gertler 1995)
- **Who responds?** Leverage (Ottonello and Winberry 2020) versus other firm characteristics eg. age and dividend status (Cloyne et al. 2023) give different answers about which balance sheets matter
- **When?** Transmission is weaker in recessions (Tenreyro and Thwaites 2016); low rates induce bank risk-taking (Jiménez et al. 2014)
- **Pass-through heterogeneity:** bank balance sheets shape how much of a policy move reaches borrowers (Altavilla et al. 2020)
- **Identification:** policy announcements convey information about the outlook, contaminating measured responses (Nakamura and Steinsson 2018; Jarociński and Karadi 2020)

The credit channel is not fully understood cont'd

- **Geography** The bulk of the evidence is from the United States and the euro area
- Transmission in emerging and low-income economies is weaker and less reliable, reflecting shallower financial markets and bank concentration (Mishra et al. 2012)
- South Africa: the bank lending channel operates but is heterogeneous (Pirozhkova and Viegi 2025); policy has distributional effects through labour and credit markets (Merrino 2022; Miyajima 2024); and regulation also shapes bank lending (Sibande et al. 2025; Sibande and Milne 2025)
- Physical climate exposure is larger and agriculture is more rainfall dependent in these economies (Schlenker and Lobell 2010), so the interaction between climate and policy is likely to be first order

Climate and the macroeconomy

- Temperature shocks lower growth, mainly in poorer countries (Dell et al. 2012)
- Weather deviations from seasonal norms are a credible source of exogenous variation (Dell et al. 2014; Hsiang 2016; Auffhammer et al. 2013)
- Disasters depress output and recovery is slow in disaster-prone economies (Noy 2009; Felbermayr and Gröschl 2014; Cantelmo et al. 2023)
- Mechanisms map directly onto the EFP: production losses, capital damage, lower collateral values, lower net worth (Dafermos et al. 2018)

Climate shocks and prices: the monetary policy nexus

- Extreme weather raises consumer prices, especially food, in developing countries (Heinen et al. 2019); disasters are inflationary in the short run (Parker 2018)
- Temperature shocks raise inflation (Mukherjee and Ouattara 2021), with asymmetric and seasonal effects (Ciccarelli et al. 2024) and persistence of about 12 months (Kotz et al. 2024)
- Central bank dilemma: a climate shock is a supply shock but can generate second-round effects and expectations risk.
  - Optimal policy under relative price shocks (Aoki 2001); EM central banks may need to react to food inflation (Anand et al. 2015)

Climate shocks, banks and credit

- After disasters, banks reallocate credit toward affected areas (Cortés and Strahan 2017), engage in recovery lending (Koetter et al. 2020), and adjust risk exposure (Schüwer et al. 2019)
- Firms hit by adverse weather draw on bank credit lines within a quarter (Brown et al. 2021)
- Banks **reprice risk**: loan spreads rise even for borrowers at risk but not directly affected (Correa et al. 2026)
- Households: insurance payouts are used to repay mortgages (Gallagher and Hartley 2017); disasters raise financial distress (Ho et al. 2023); mortgage standards tighten in disaster-prone areas (Duanmu et al. 2022)
- Climate risk is a financial stability issue (Dafermos et al. 2018; Giglio et al. 2021; Hong et al. 2020)

The gap this paper addresses

- **Literature A:** monetary policy → credit through the EFP, identified with policy surprises
- **Literature B:** climate shocks → balance sheets, loan pricing, inflation
- **Gap:** does the climate state change the **strength** of the credit channel?
- Our contribution:
  - Bank-level, product-level lending rates and volumes for South Africa, 2011 to 2025
  - Three identified policy surprises × two population-weighted climate shocks
  - Fixed effects, panel local projections and asymmetric (extreme-state) local projections

Why surprises rather than the policy rate

- The repo rate r_t is **endogenous**: it responds to inflation, output and, in a climate-exposed economy, potentially to food prices and the weather
- Regressing credit on r_t confounds policy with the conditions that prompted policy
- We need variation in policy that is orthogonal to the policymaker's and the market's information sets (Romer and Romer 2004)
- Two families of surprise measures:
  - **Structural / forecast based:** residual of the rate change on forecasts (Romer and Romer; Miyajima)
  - **High frequency / market based:** asset price moves in a narrow window around announcements (Kuttner 2001; Gürkaynak et al. 2005)

Structural surprises (i): the Romer and Romer measure

Following Romer and Romer (2004), adapted to South Africa by Merrino (2022):

$$\Delta r_t = \alpha + \tilde{r}_t + \sum_{j=0}^{2} \beta_j \Delta \tilde{y}_{t+j} + \sum_{j=0}^{2} \delta_j \tilde{\pi}_{t+j} + \varepsilon_t$$

- Δr_t: change in the repo rate forecast; $\tilde{r}_t$, $\tilde{y}_t$, $\tilde{\pi}_t$: forecasts of the repo rate, real GDP growth and inflation
- Forecasts up to two quarters ahead capture the common assumption of unchanged policy in the near term
- The residual ε_t is the component of the policy change not explained by what policymakers already expected: MPShock_t^{Romer}

Structural surprises (ii): the Miyajima measure

Following Miyajima (2024), built from Bloomberg consensus forecast errors:

$$FE_t^i = EXP_t^i - ACT_t^i, \qquad i \in \{\text{policy rate}, \pi, y\}$$

$$FE_t^r = \alpha + \beta\, FE_t^\pi + \gamma\, FE_t^y + \epsilon_t$$

- The policy rate forecast error is purged of inflation and growth news
- The residual ϵ_t is the part of the policy miss that markets could not have anticipated from macro news: $\text{MPShock}_t^{Miyajima}$
- Advantage: uses market participants' expectations rather than the central bank's own projections

High-frequency surprises

- Measure asset price changes in a narrow window around each MPC announcement (Kuttner 2001; Gürkaynak et al. 2005)
- Because little else happens in the window, the change in financial market prices is attributed to policy news (Nakamura and Steinsson 2018)
- For South Africa, Pirozhkova and Viegi (2025) extract target, forward guidance, central bank information and country risk factors from market prices around MPC dates
- We use the **target factor** as our third surprise measure. The target factor is the component of financial-market movements around MPC announcements that captures unexpected changes in the current policy stance.

Surprises do not transmit like textbook policy

- Kuttner (2001): the **unanticipated** component of a rate change moves market rates

$$\Delta i_t = \alpha + \beta_e \, \Delta r_t^e + \beta_u \, \Delta r_t^u + \eta_t, \qquad \hat{\beta}_e \approx 0, \;\; \hat{\beta}_u \gg 0$$

- Credit costs respond far more to surprises than a pure expectations model predicts (Gertler and Karadi 2015)
- Announcements also carry information about the outlook, so some "tightening" surprises can actually lower spreads (Nakamura and Steinsson 2018; Jarociński and Karadi 2020)
- Lending-rate pass-through is slower when a move is **unexpected** (Kleimeier and Sander 2006)
- Takeaway: an unanticipated move is exactly where EFP frictions have the most room to bite, which is part of why we focus on surprises

The three surprise series

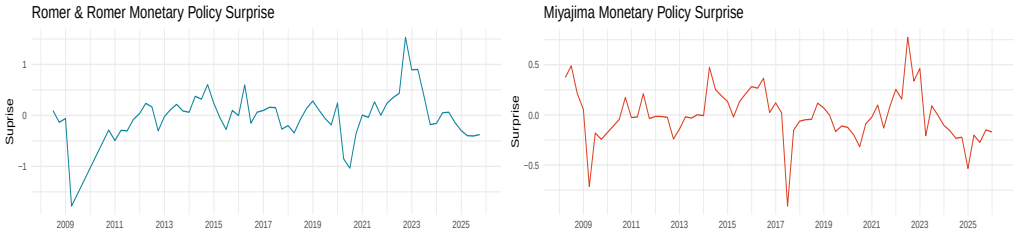
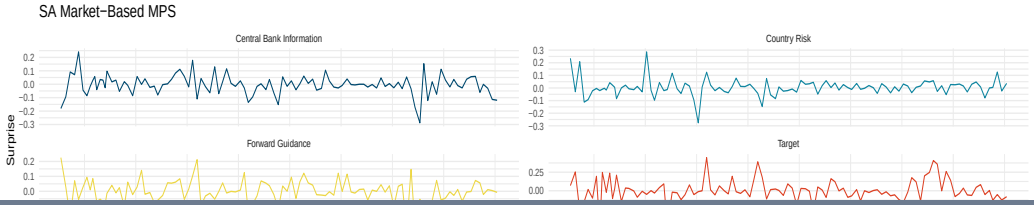


Figure 1



Climate shocks as exogenous variation

- We use deviations of population-weighted temperature and precipitation from their calendar-month norms, in standard deviation units:

$$\text{ClimateShock}_t = \frac{T_t - \overline{T}_t}{\sigma(T_t)}$$

- $\overline{T}_t$ and $\sigma(T_t)$ are the calendar-month mean and standard deviation over 2011 to 2025, so seasonality is removed by construction
- Weather realisations are plausibly exogenous to credit conditions and to policy within the quarter (Dell et al. 2014; Hsiang 2016)
- Source: ERA5 reanalysis, aggregated over South Africa with population weights (Auffhammer et al. 2013)

The climate shocks series

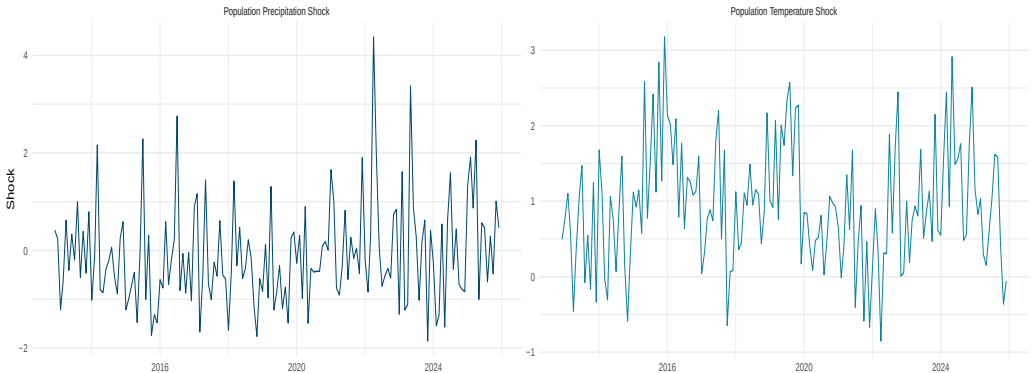


Figure 3

Positive values indicate warmer or wetter than the historical monthly norm; negative values cooler or drier. The 2015/16 El Niño drought and the 2022 flooding episodes are visible as tail observations.

Conceptual framework (i): corporates

Following the external finance premium logic (Bernanke et al. 1996; Laumer and Schaffer 2025):

$$R_{it}^{c} = R_t + f(D_t/E_t)\,, \qquad f' > 0$$

- A climate shock disrupts supply chains, damages capital and destroys output: revenues and profits fall
- Collateral values fall; debt D_t rises; equity E_t falls $\Rightarrow$ $f(D_t/E_t)$ rises
- **Amplification:** the climate shock raises the premium at the same time as a tightening raises R_t
- **Dampening:** firms with cash buffers or insurance see no change in D_t/E_t (Sharpe and Suarez 2020)
- Another, related transmission avenue, is firms that must rebuild or adapt borrow even at higher rates (Xu et al. 2025)

Conceptual framework (ii): households

$$R_{it}^{h} = R_t + \phi_t, \qquad \phi_t > 0$$

- ϕ_t: household-specific spread reflecting creditworthiness, indebtedness, income and collateral; transmission differs across households by income exposure (Slacalek et al. 2020)
- **Amplification:** lower income and wealth and damaged collateral raise ϕ_t; affected households pay more for a given policy rate
- **Dampening:** liquid assets and insurance absorb losses without widening ϕ_t; insured households use payouts to repay rather than borrow (Gallagher and Hartley 2017)

Conceptual framework (iii): banks

- Banks connect the two: climate episodes raise the weight lenders put on balance sheet strength due to increased arrears and financial distress (Ho et al. 2023; Duanmu et al. 2022; Correa et al. 2026)
- Weaker loan demand after a shock can instead leave banks with less scope to reprice, dampening the rate response
- Whether the net effect is amplification or dampening is therefore an empirical question
- We test the net effect without attributing it to a single mechanism

Data

- **Sample:** South Africa, 2011Q1 to 2025Q1, quarterly; bank-quarter panel of the major banking groups
- **Credit (SARB BA900 and BA930 returns, bank level):** lending rates (percentage points) and log credit volumes for corporate and household borrowers, each split into unsecured, secured and mortgage credit, constructed following Sibande et al. (2025) and Sibande and Milne (2025)
- **Monetary policy surprises:** Romer and Romer (SARB forecasts), Miyajima (Bloomberg consensus forecast errors), Target factor (Pirozhkova and Viegi 2025); all aggregated to quarters
- **Climate shocks:** ERA5 population-weighted temperature and precipitation deviations from calendar-month norms
- **Controls (robustness):** GDP growth, USD/ZAR, unemployment rate

Bank-level lending rates by borrower and product

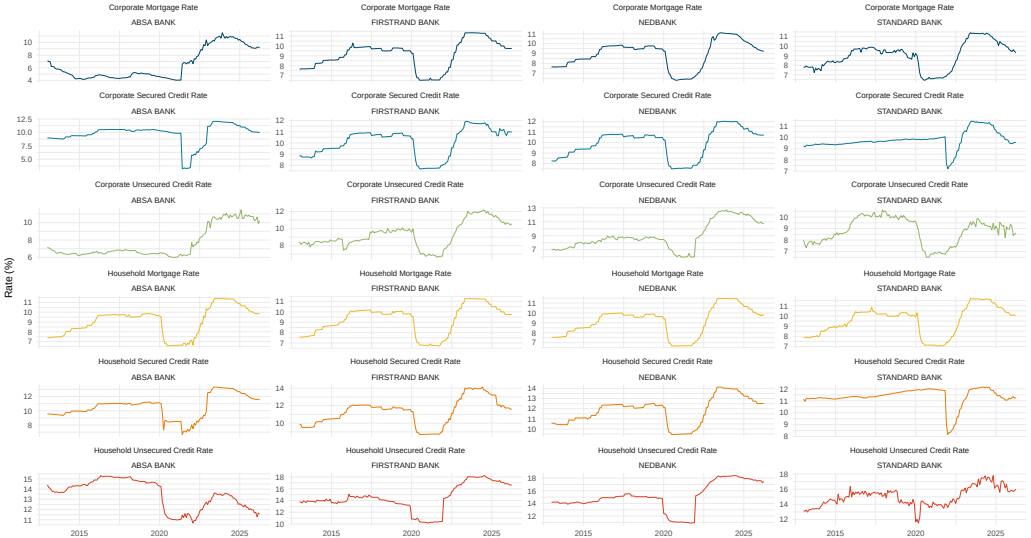


Figure 4

Contemporaneous specification: panel fixed effects

$$\begin{aligned}\text{Credit}_{s,t} = \beta_0 + \beta_1\, \text{CS}_t^k + \beta_2\, \text{MP}_t + \beta_3\, (\text{CS}_t^k \times \text{MP}_t) \\ + \Gamma X_t + \gamma_s + \tau t + \varepsilon_{s,t}\end{aligned}$$

- CS_t^k: climate shock k (temperature or precipitation); MP_t: monetary policy surprise
- $\text{Credit}_{s,t}$: lending rate or log volume for bank s and a given product
- γ_s: bank fixed effects; τt: linear trend; quarter-of-year effects absorb seasonality
- β_2: effect of a surprise tightening without a climate shock
- β_3: **the coefficient of interest.** $\beta_3 \neq 0$ means the climate state changes the strength of transmission; the sign relative to β_2 tells us amplification versus dampening
- Standard errors clustered at the bank level

Dynamic specification: panel local projections

For each horizon $h = 0, 1, \ldots, 12$ quarters (Jordà 2005):

$$\begin{aligned}\text{Credit}_{s,t+h} - \text{Credit}_{s,t-1} = \beta_0^h + \beta_1^h \, \text{CS}_t^k + \beta_2^h \, \text{MP}_t + \beta_3^h \, (\text{CS}_t^k \times \text{MP}_t) \\ + \Gamma X_t + \gamma_s + \tau t + \varepsilon_{s,t}^h\end{aligned}$$

- Impulse responses: $\widehat{\beta}_2^h$ (policy alone, blue) and $\widehat{\beta}_2^h + \widehat{\beta}_3^h$ (policy with a one standard deviation climate shock, red)
- The vertical gap between the two paths at each horizon is the climate amplification or dampening of transmission
- Driscoll-Kraay standard errors for cross-sectional and serial dependence (Driscoll and Kraay 1998); 68% and 90% bands

Extreme states: asymmetric local projections

- Average interactions can mask what happens in the tails, which is where the conceptual framework predicts the largest balance sheet damage
- We evaluate the dynamic response at **extreme climate states**:

$$\text{IRF}^h(c) \;=\; \hat{\beta}_2^h + \hat{\beta}_3^h \cdot c, \qquad c \in \{+3, -3\}$$

  - $c = +3$: extreme heat
  - $c = -3$: extreme lack of precipitation

What we mean by amplification and dampening

- **Policy alone:** a one unit surprise tightening moves the lending rate by β_2
- **Policy during a climate shock:** the same tightening moves it by $\beta_2 + \beta_3$

Case	What happens to the policy effect	Rule
Amplification	Same direction, **bigger**: e.g. up 70 bp instead of 50 bp. The climate shock pushes on the same lever (a higher premium), so policy gets more traction	$\lvert\beta_2 + \beta_3\rvert > \lvert\beta_2\rvert$
Dampening	Same direction, **smaller**: e.g. up 30 bp instead of 50 bp. Buffers, insurance or weak loan demand mean the premium does not widen, so policy loses traction	$\lvert\beta_2 + \beta_3\rvert < \lvert\beta_2\rvert$

- Note what this is **not**: it is not the direct effect of the weather on credit (β_1).

Contemporaneous evidence: lending rates

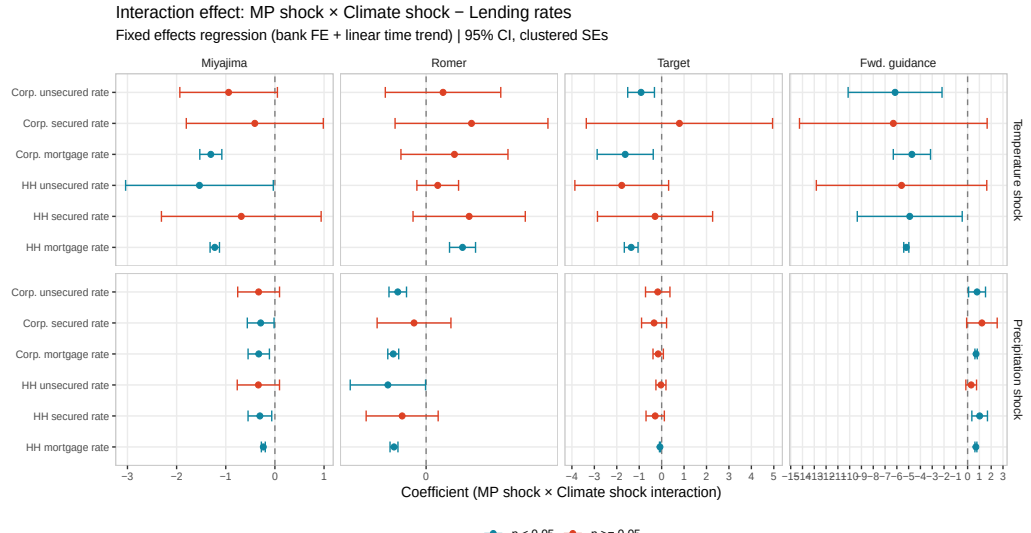


Figure 5

Contemporaneous evidence: credit volumes

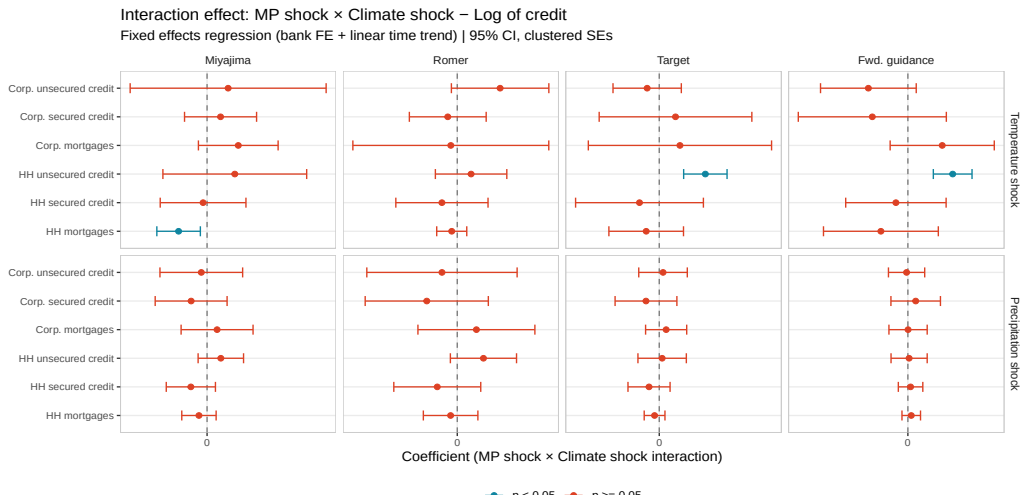


Figure 6

Corporate lending rates: temperature shock interaction

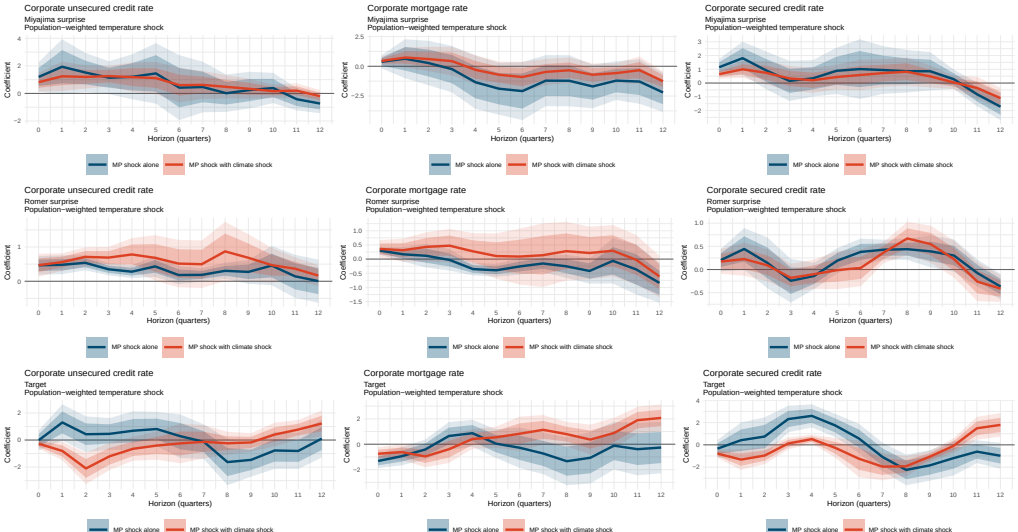


Figure 7

Household lending rates: temperature shock interaction

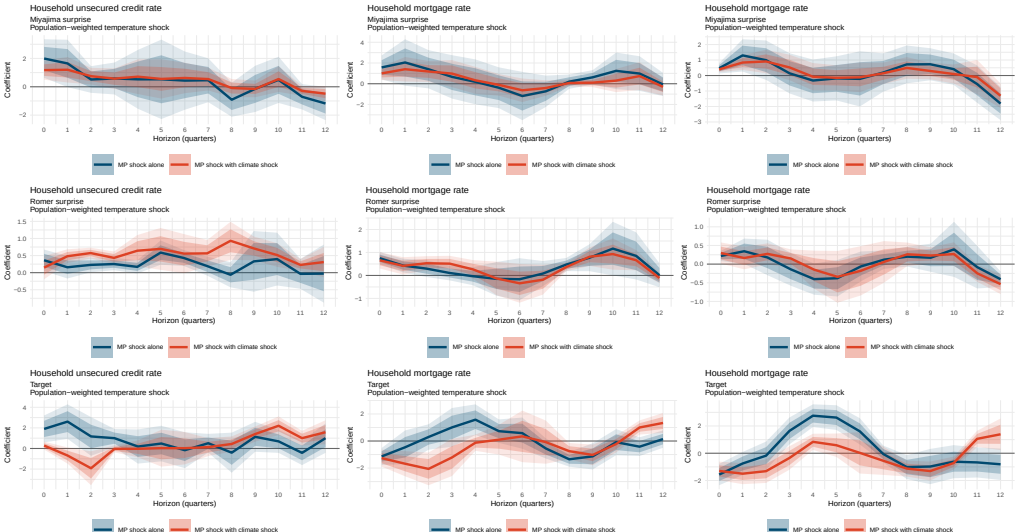


Figure 8

Corporate lending rates: precipitation shock interaction

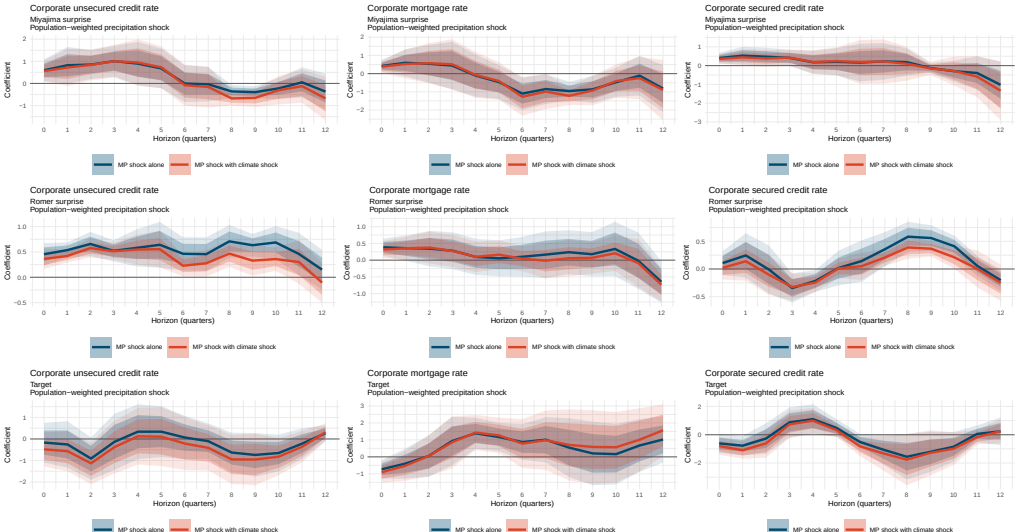


Figure 9

Household lending rates: precipitation shock interaction

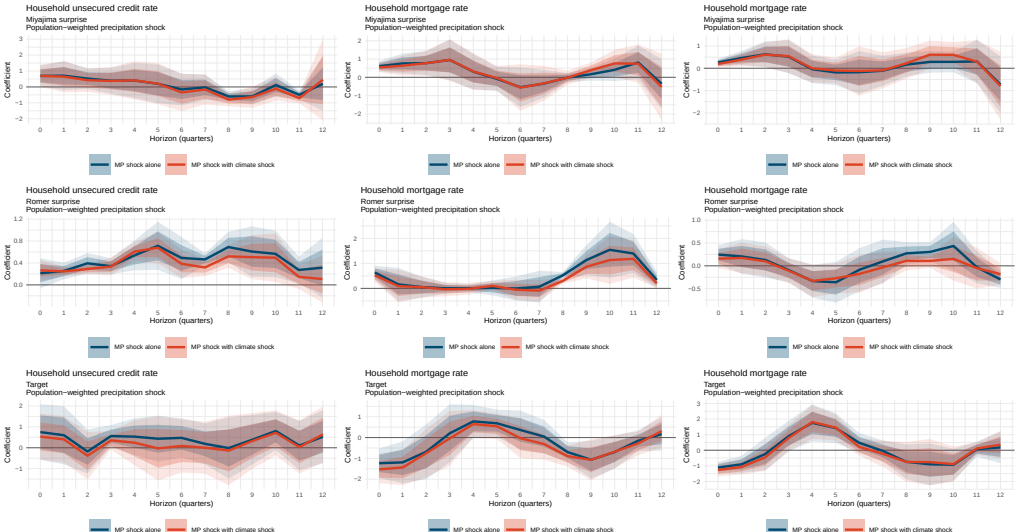


Figure 10

Reading the responses: rates are held higher for longer

- Policy alone: lending rates rise after a surprise tightening and typically **revert** within 6 to 10 quarters (blue paths)
- With a temperature shock: there is some evidence that the lending rate path stays **elevated** beyond the point at which the policy-only response has faded (red paths)
- Interpretation through the framework: the tightening raises R_t, the heat shock raises $f(D_t/E_t)$ and ϕ_t, and the premium decays more slowly than the policy impulse itself
- The **price** of credit is what adjusts. Volume responses (appendix) show little systematic interaction, consistent with banks repricing risk rather than rationing quantities (Correa et al. 2026; Altavilla et al. 2020)
- The most compelling evidence exists for the case of temperature shocks warranting further investigation into hot/dry extreme cases

A real-world episode: the 2015/16 El Niño drought in South Africa

MPC date	Decision	Drought-related rationale
Nov 2015	+25 bp to 6.25%	Lagged effect of severe drought on food prices flagged
Jan 2016	+50 bp to 6.75%	"the impact of the worsening drought on food prices is becoming increasingly evident"; food inflation seen peaking near 11%
Mar 2016	+25 bp to 7.00%	Food price pressures from the drought intensified by more than forecast; headline above the 3 to 6% target
Mar 2016 to Jul 2017	**Held at 7.00%** for 16 months	Food inflation in double digits in late 2016; first cut only in July 2017

- This episode shows the mechanism through the policy rate; our estimates show it continues at the **lending rate**, through the premium
- Prime lending rates moved one for one to 10.5% and stayed there for the whole period: the price of credit was held higher for longer in the presence of a severe climate shock

Direction of the interaction across products and surprises

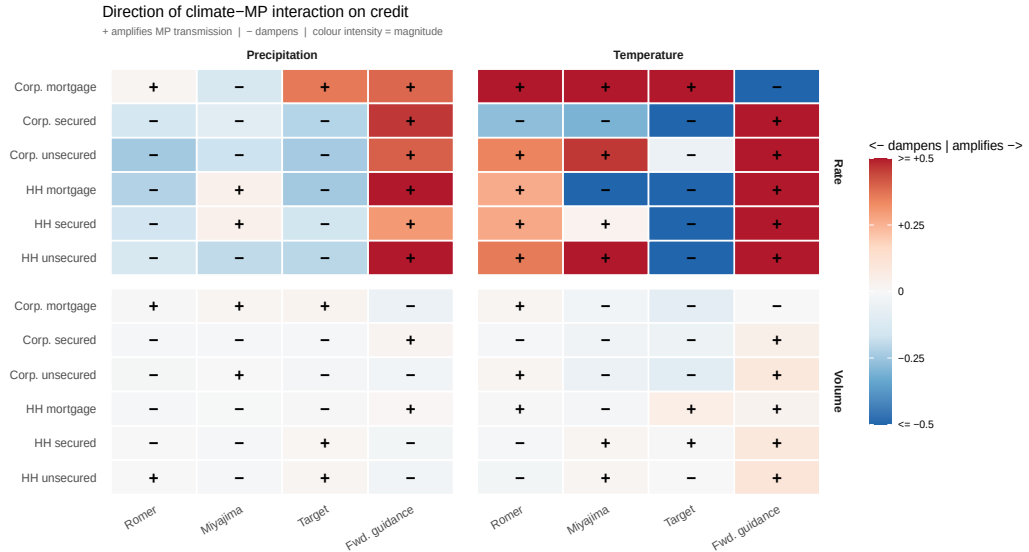


Figure 11

Precipitation: a threshold story

- The heatmap shows **mixed signs** for precipitation but mostly **amplification** for temperature in lending rates
- Consistent with the framework: a one standard deviation precipitation shock in a rainfed economy is often **beneficial** for agricultural and downstream income, which lowers the premium and **dampens** the rate response
- Temperature shocks are almost uniformly adverse in a warm climate (the economy is beyond the optimum of Burke et al. (2015)), so they amplify lending rates

Extreme heat: corporate lending rates

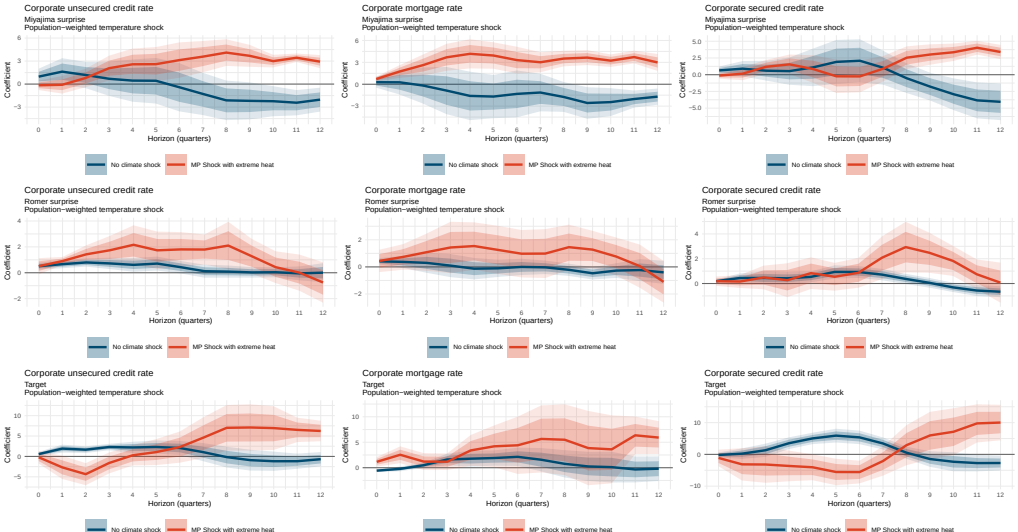


Figure 12

Extreme heat: household lending rates



Figure 13

Extreme lack of precipitation: corporate lending rates

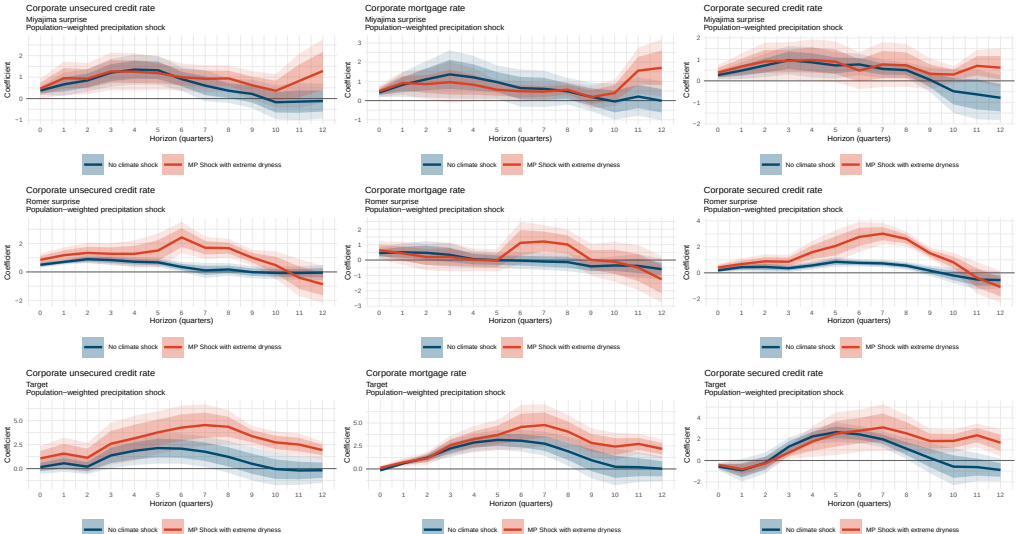


Figure 14

Extreme lack of precipitation: household lending rates

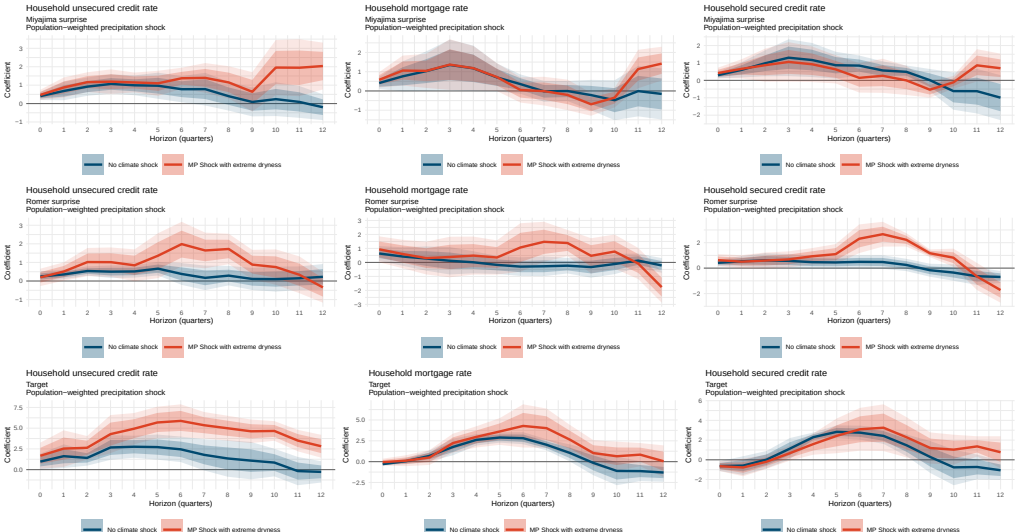


Figure 15

What the extreme-state estimates show

- Under extreme heat, the lending rate paths **diverge sharply** from the no-climate baseline: for most corporate and household products the red path rises and stays positive through 12 quarters while the blue path reverts or turns negative
- Under an extreme lack of precipitation, the same pattern appears, now with precipitation unambiguously adverse: the threshold story is confirmed at the tail
- The gap between the paths is the marginal climate effect at that intensity: economically large, and generally significant
- Tail risk interpretation: moderate seasonal variability does little to transmission; **severe and unusual** episodes create the frictions that keep the price of credit high

Persistence and timing of the interaction

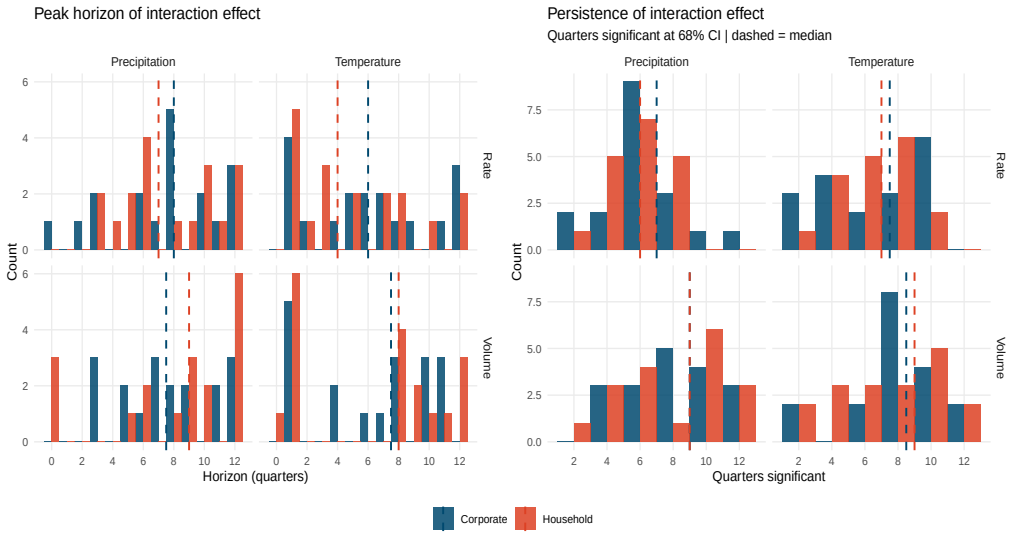


Figure 16

Persistence by credit product

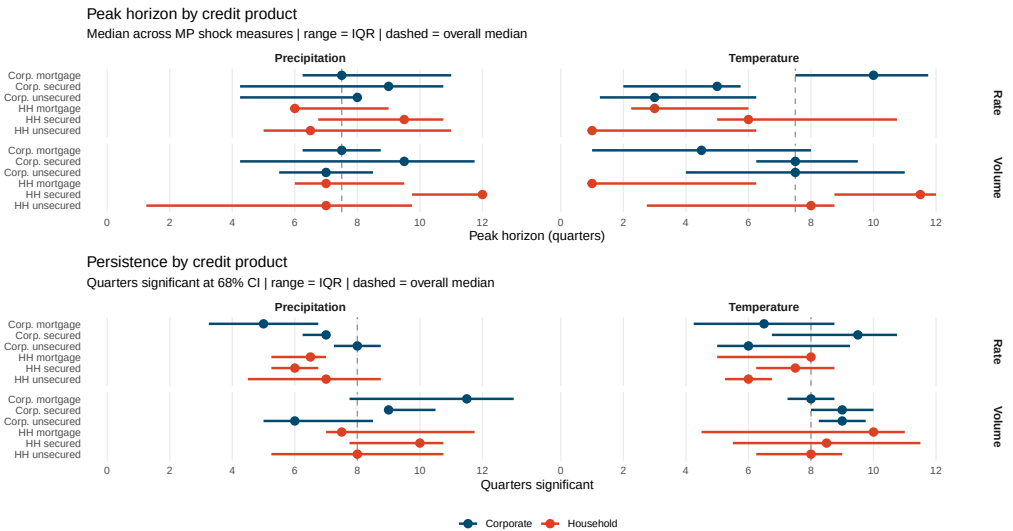


Figure 17

How long does the climate effect last?

- The **peak** of the absolute interaction effect occurs at a median of roughly **8 quarters** after the shock, with a mode between 8 and 10 quarters for lending rates
- Persistence of two years is consistent with: (i) the slow repair of net worth and collateral after physical damage, (ii) persistent inflationary effects of heat (Kotz et al. 2024), and (iii) sluggish and asymmetric lending-rate adjustment (Altavilla et al. 2020; Kleimeier and Sander 2006)

Summary of findings

1. **Transmission is augmented by climate shocks.** The interaction between policy surprises and climate shocks is significant across a large share of product-horizon combinations
2. **The evidence is in prices, not quantities.** Significant interactions concentrate in lending rates; log volumes show little consistent interaction. This is risk repricing through the external finance premium, not rationing
3. **Temperature amplifies; precipitation depends on the threshold.** Heat keeps lending rates higher for longer; moderate rain can dampen, drought amplifies
4. **The effect is a tail phenomenon.** Extreme-state responses diverge sharply and durably from the baseline
5. **It is persistent.** Interaction peaks around 8 quarters and remains significant for roughly two years

Policy implications

- **Disentangle climate from other shocks.**
- **Unintended restrictiveness.** A tightening that coincides with a severe heat or drought episode transmits more strongly and for longer through lending rates than the policy rate path implies. Models that ignore the climate state will overstate the room for tightening and understate how long the restriction lasts
- **Room for a more targeted response.**
- **Climate-vulnerable central banks** would benefit from incorporating physical climate risk into their transmission models, consistent with the financial-stability view of climate (Dafermos et al. 2018; Giglio et al. 2021)

Limitations

- No bank-level **sectoral** credit data: cannot trace the interaction to agriculture, energy or construction lending
- **Quarterly** frequency limits the precision of impact timing; monthly rates would sharpen the impact horizon
- Surprise measures are standard; measures that capture the SARB's climate-related communication, or that separate physical from financial-stability news, may reveal further dimensions

Thank you